

补充作业1

Q1

1. Suppose a sequence $\{a_n\}$ is bounded and $\lim_{n \rightarrow \infty} b_n = 0$. Prove that $\lim_{n \rightarrow \infty} a_n b_n = 0$.

$$\text{pf: } -M|b_n| \leq |a_n b_n| \leq M|b_n|$$

$$\begin{array}{ccc} \downarrow & & \downarrow \\ 0 & & 0 \end{array} \quad n \rightarrow \infty$$

By sandwich Thm $\lim_{n \rightarrow \infty} |a_n b_n| = 0 \Rightarrow \lim_{n \rightarrow \infty} a_n b_n = 0$.

□

Q2

2. Find the following limits.

(a) $\lim_{n \rightarrow \infty} \frac{1}{n} \sin n \cos n^2$. (b) $\lim_{n \rightarrow \infty} (a^n + b^n)^{\frac{1}{n}}$ ($a > 0, b > 0$).

(c) $\lim_{n \rightarrow \infty} \frac{n^c}{a^n}$ ($a > 1, c > 0$). (d) $\lim_{n \rightarrow \infty} (1 + \frac{1}{2} + \dots + \frac{1}{n})^{\frac{1}{n}}$.

(e) $\lim_{n \rightarrow \infty} (\frac{n}{n+c})^n$ (any c) (f) $\lim_{n \rightarrow \infty} (\sqrt{n + \sqrt{n + 2\sqrt{n}}} - \sqrt{n})$.

Sol: (a)

$$-\frac{1}{n} \leq \frac{1}{n} \sin n \cos n^2 \leq \frac{1}{n}$$

$$\begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ 0 & 0 & 0 \end{array} \quad n \rightarrow \infty$$

(b) $\lim_{n \rightarrow \infty} (a^n + b^n)^{\frac{1}{n}} = \max\{a, b\}$

reason: ① $a=b$, then $\lim_{n \rightarrow \infty} (a^n + b^n)^{\frac{1}{n}} = \lim_{n \rightarrow \infty} 2^{\frac{1}{n}} a = a$

② assume $a > b$, then

$$(a^n + b^n)^{\frac{1}{n}} = a \left[1 + \frac{b^n}{a^n} \right]^{\frac{1}{n}} \cdot \left(\frac{b}{a} \right)^{\frac{1}{n}} = a e^0 = a$$

$\downarrow$ $\downarrow$ $\downarrow$
 $n \rightarrow \infty$ e 0

(c) $\lim_{n \rightarrow \infty} \frac{n^c}{a^n} = 0$

$$\therefore \lim_{x \rightarrow \infty} \frac{x^c}{a^x} \stackrel{\frac{\infty}{\infty}}{=} \lim_{x \rightarrow \infty} \frac{c x^{c-1}}{a^x \cdot \ln a} \stackrel{\frac{\infty}{\infty}}{=} \dots = 0.$$

(d)

$$1^{\frac{1}{n}} \leq \left(1 + \frac{1}{2} + \dots + \frac{1}{n} \right)^{\frac{1}{n}} \leq n^{\frac{1}{n}}$$

$\downarrow$ $\downarrow$ $\downarrow$ $n \rightarrow \infty$
 1 1 1

(e)

$$\left(\frac{n}{n+c} \right)^n = \left(1 + \frac{-c}{n+c} \right)^n = \left(1 + \frac{-c}{n+c} \right)^{\frac{n+c}{-c} \cdot \frac{-cn}{n+c}}$$

$\downarrow$ $\downarrow$ $\downarrow$
 $n \rightarrow \infty$ e $-c$

$\therefore$ 原极限为 e^{-c}

$$\begin{aligned}
 (f) \quad \sqrt{n + \sqrt{n + 2\sqrt{n}}} - \sqrt{n} &= \frac{n + \sqrt{n + 2\sqrt{n}} - n}{\sqrt{n + \sqrt{n + 2\sqrt{n}}} + \sqrt{n}} \\
 &= \frac{\sqrt{1 + \frac{2}{\sqrt{n}}}}{\sqrt{1 + \sqrt{\frac{n + 2\sqrt{n}}{n^2}}} + 1} \\
 &\quad \downarrow \\
 &= \frac{\sqrt{1 + 0}}{\sqrt{1 + 0} + 1} = \frac{1}{2}
 \end{aligned}$$

□

Q3

3. Determine if the following sequences converge or not and give your reasons.
If so, find their limit.

(a) $a_n = \sin n$. (b) $a_n = n^{(-1)^n}$. (c) $a_n = \cos(\sqrt{4n^2 + 1}\pi)$.

Sol.: (a) diverge: assume $\lim_{n \rightarrow \infty} \sin n = a$, then $\lim_{n \rightarrow \infty} \sin(n+2) = a$, then

$$0 = \lim_{n \rightarrow \infty} \sin(n+2) - \sin n = \lim_{n \rightarrow \infty} 2 \cos(n+1) \sin 1, \text{ then } \lim_{n \rightarrow \infty} \cos n = \lim_{n \rightarrow \infty} \cos(n+1) = 0,$$

$$\text{then } \lim_{n \rightarrow \infty} \sin n = \lim_{n \rightarrow \infty} \sqrt{1 - \cos^2 n} = 1, \text{ but } 1 = \lim_{n \rightarrow \infty} \sin 2n = \lim_{n \rightarrow \infty} 2 \sin n \cos n = 0$$

contradiction!

(b) diverge $1, 2, \frac{1}{3}, 4, \frac{1}{5}, 6, \dots$

(c) converge

$$\begin{aligned}
 \cos(\sqrt{4n^2 + 1}\pi) &= \cos(\sqrt{4n^2 + 1}\pi - 2n\pi) \\
 &= \cos\left(\frac{\pi}{\sqrt{4n^2 + 1} + 2n}\right)
 \end{aligned}$$

□

Q4

4. Let $a_1 \in (0, \frac{\pi}{2})$ and $a_{n+1} = \sin a_n$. Does the sequence $\{a_n\}$ converge? Why? If so, find its limit.

Sol: Converge, $a = \lim_{n \rightarrow \infty} a_{n+1} = \lim_{n \rightarrow \infty} \sin a_n$, then $a = \sin a$, thus $a = 0$

$$\begin{cases} \text{if } a_n \in (0, \frac{\pi}{2}) \Rightarrow a_{n+1} = \sin a_n > 0 \\ a_{n+1} - a_n = \sin a_n - a_n \leq 0 \end{cases}$$

$\Rightarrow$ If $a_n \in (0, \frac{\pi}{2})$, we have $0 < a_{n+1} \leq a_n < \frac{\pi}{2}$.

then since $a_1 \in (0, \frac{\pi}{2})$, we have $0 < a_n < \frac{\pi}{2}$ for all n and $\{a_n\}$ monotone decrease.

Therefore by monotonic sequence Thm, $\{a_n\}$ converge.

□

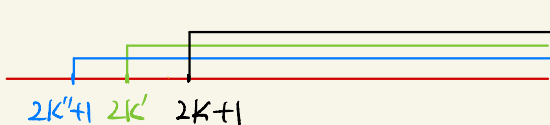
Q5.

133. For a sequence $\{a_n\}$ the terms of even index are denoted by a_{2k} and the terms of odd index by a_{2k+1} . Prove that if $a_{2k} \rightarrow L$ and $a_{2k+1} \rightarrow L$, then $a_n \rightarrow L$.

#: $a_{2k} \rightarrow L$ means: $\forall \varepsilon > 0, \exists K' \in \mathbb{N}$, s.t. w $k > K'$, we have $|a_{2k} - L| < \varepsilon$

$a_{2k+1} \rightarrow L$ means: $\forall \varepsilon > 0, \exists K'' \in \mathbb{N}$, s.t. w $k > K''$, we have $|a_{2k+1} - L| < \varepsilon$

$\therefore$ for $\forall \varepsilon > 0, \exists K = \max\{K', K''\}$ as above, s.t. w $n > 2K+1$, we have $|a_n - L| < \varepsilon$, which means $a_n \rightarrow L$.



$$\begin{aligned} |a_{2k} - L| &< \varepsilon \\ |a_{2k+1} - L| &< \varepsilon \\ |a_n - L| &< \varepsilon \end{aligned}$$

□